

Miguel Tomás Olórtogui Huamán

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Mechatronics Engineer from UTEC (top 10% of the class, specialization in robotics), currently pursuing a Master of Science in Engineering with research on learning from demonstration for humanoid robots. Experience with robotic manipulators (UR5, Sawyer), humanoid robots (Unitree H1-2), control, computer vision and embedded systems, and in teaching as a Senior Teaching Assistant and laboratory instructor.

EDUCATION

Universidad de Ingeniería y Tecnología (UTEC) | Graduate School 2026 – Present
Master of Science in Engineering (Maestría en Ciencias de la Ingeniería) – in progress.

- Thesis: *Development of a learning-from-demonstration algorithm for robotic assistance in fruit picking on a conveyor belt using computer vision and a humanoid robot.* Advisor: Dr. Ruth Canahuire.

Universidad de Ingeniería y Tecnología (UTEC) 2019 – 2024
Bachelor's degree in Mechatronics Engineering – top 10%, specialization in robotics. Thesis: wall-climbing robot control.

TEACHING AND PROFESSIONAL EXPERIENCE

UTEC | Laboratory Instructor – Introduction to Mechatronics Engineering 2026 – Present
- Lead the course laboratory sessions and advise first-year students on hands-on mechatronics projects.

UTEC | Senior Teaching Assistant 2026 – Present
- Support courses of the Mechatronics Engineering program and mentor students in labs and projects.

UTEC | Teaching and Lab Assistant March 2023 – 2025
- Assisted in the Digital Circuits course, guiding students in programming FPGA devices (Basys 3 board).
- Supported the Automatic Control course: Quanser equipment and PLC configuration with TIA Portal.

UTEC | Robotics Lab Intern January 2023 – 2025
- Collaborated on laboratory research, maintained robotic equipment and gave robotics outreach workshops.

RESEARCH AND PROJECTS

Learning from Demonstration for Fruit Picking with the Unitree H1-2 Humanoid (Master's thesis) 2026 – Present

- Built a markerless motion-capture system with three calibrated RGB cameras (ChArUco, MediaPipe Pose, multi-view triangulation) with about 1 mm of metrological error.
- Developed whole-body inverse kinematics with joint limits (97% usable solutions vs. 13%) and a three-layer bimanual collision-avoidance controller (QP with velocity dampers, arm coordination, RRT-Connect) in ROS 2 and MuJoCo.
- RGB-D fruit localization by sphere fitting (~2.4 mm error) and visual servoing: 3/3 grasps on a moving belt in sim.

Humanoid Line Follower with the Unitree H1-2 (Robotics40 challenge) 2026

- Line following with only the head camera: shadow-robust IR perception, pure pursuit, supervisor; 12/12 sim runs.

UR5 Painter Robot (team project) 2026 – Present

- ROS 2 pipeline that turns text and images into UR5 drawing strokes, with force-based calibration of the surface.

SwarmBot: Swarm of Mobile Robots with micro-ROS 2026 – Present

- Differential-drive robots on ESP32-S3 (ESP-IDF, FreeRTOS, micro-ROS) connected to ROS 2 over Wi-Fi: IMU publishing, 100 Hz wheel PI control and automatic reconnection.

Automated Beverage System with UR5 Manipulator Arm (Robot Barista) September 2023 – June 2024

- Designed the UR5 control algorithm to serve beverages; integrated a Siemens S7-1500 PLC with an HMI.

Sawyer Robot: Vision-Based Pick-and-Place and Tele-Surgery Motion March 2024 – December 2024

- Object classification and trajectory generation from camera images; hierarchical controller that keeps the end-effector orientation of a redundant robot, simulated in Gazebo.

LEADERSHIP EXPERIENCE

Spokesperson and Representative for Mechatronics Engineering at UTEC January 2022 – April 2023
- Represented the program at academic events, linking students, faculty and administrative staff.

Chair of IEEE CAS UTEC (2022) and Secretary of IEEE UTEC (2023) 2022 – 2023
- Led the chapter and coordinated more than 10 technical events in 2022; organized 5 inter-university events in 2023.

SKILLS AND LANGUAGES

- **Programming and vision:** Python, C++, C, MATLAB/Simulink; OpenCV, MediaPipe, YOLO, camera calibration.
- **Robotics:** ROS / ROS 2, MuJoCo, Gazebo, micro-ROS; kinematics, nonlinear and QP control, motion planning.
- **Embedded and automation:** ESP32 (ESP-IDF, FreeRTOS), STM32, Raspberry Pi, Arduino, Siemens PLC, FPGA.
- **Languages:** Spanish (native); English (advanced, Asociación Cultural Peruano Británica).